

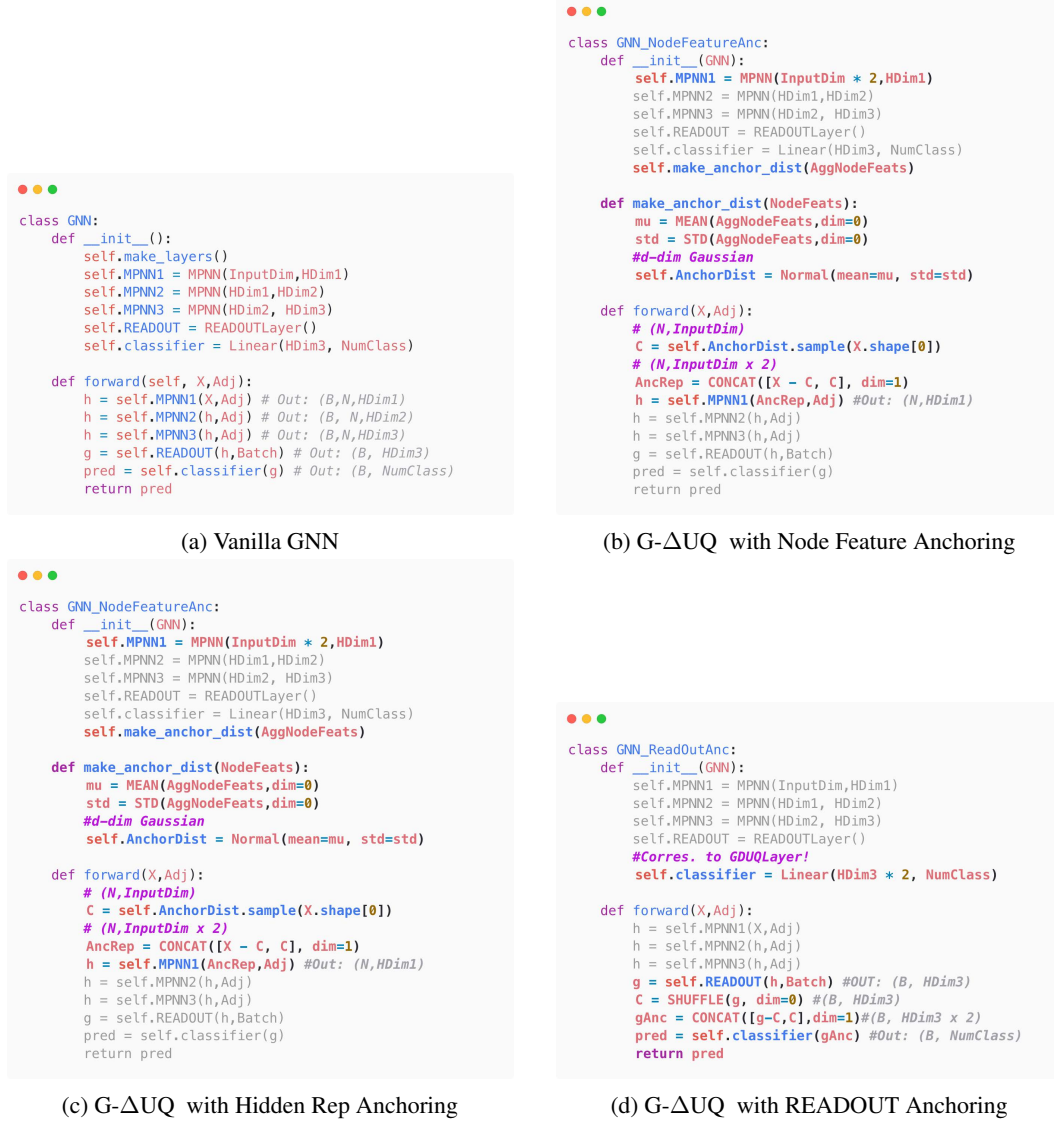
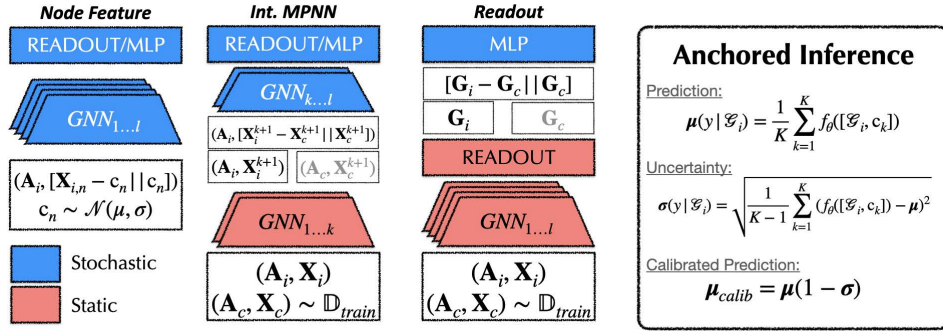
A.1 PSEUDOCODE FOR G- Δ UQ

Figure 5: **PseudoCode for G- Δ UQ**. We provide simplified pseudo-code to demonstrate how anchoring can be performed. We assume PyTorchGeometric style mini-batching. Changes with respect to the vanilla GNN are shown in bold. Unchanged lines are grayed out.



Overview of G- Δ UQ models. Here, we present a conceptual overview of how G- Δ UQ induces partially stochastic models. This figure is complementary to 5.

A.2 ETHICS STATEMENT

This work proposes a method to improve uncertainty estimation in graph neural networks, which has potential broader societal impacts. As graph learning models are increasingly deployed in real-world applications like healthcare, finance, and transportation, it becomes crucial to ensure these models make reliable predictions and know when they may be wrong. Unreliable models can lead to harmful outcomes if deployed carelessly. By improving uncertainty quantification, our work contributes towards trustworthy graph AI systems.

We also consider several additional safety-critical tasks, including generalization gap prediction for graph classification (to the best of our knowledge, we are the first to report results on this task) and OOD detection. We hope our work will encourage further study in these important areas.

However, there are some limitations. Our method requires (modest) additional computation during training and inference, which increases resource usage. Although G- Δ UQ, unlike post-hoc methods, does not need to be fit on a validation dataset, evaluation of its benefits also relies on having some out-of-distribution or shifted data available, which may not always be feasible. We have seen in Table 1 that there are tasks for which G- Δ UQ fails to improve accuracy and/or calibration of some post-hoc methods, further emphasizing the need to perform appropriate model selection and the risks if shifted validation data is not available. Finally, there are open questions around how much enhancement in uncertainty calibration translates to real-world safety and performance gains.

Looking ahead, we believe improving uncertainty estimates is an important direction for graph neural networks and deep learning more broadly. This will enable the development safe, reliable AI that benefits society. We hope our work inspires more research in the graph domain that focuses on uncertainty quantification and techniques that provide guarantees about model behavior, especially for safety-critical applications. Continued progress will require interdisciplinary collaboration between graph machine learning researchers and domain experts in areas where models are deployed.

A.3 REPRODUCIBILITY

For reproducing our experiments, we have made our code available at this anonymous [repository](#). In the remainder of this appendix (specifically App. A.6, A.7), and A.10), we also provide additional details about the benchmarks and experimental setup.